

Python List Methods

Important list methods with simple examples and outputs

```
numbers = [10, 20, 30, 40]
```

1. append()

Adds one element to the end of the list.

Example:

```
numbers = [10, 20, 30]
numbers.append(40)
print(numbers)
```

Output:

```
[10, 20, 30, 40]
```

2. extend()

Adds all elements from another iterable to the end of the list.

Example:

```
numbers = [10, 20]
numbers.extend([30, 40, 50])
print(numbers)
```

Output:

```
[10, 20, 30, 40, 50]
```

3. insert()

Inserts an element at a specified index.

Example:

```
numbers = [10, 20, 40]
numbers.insert(2, 30)
print(numbers)
```

Output:

```
[10, 20, 30, 40]
```

4. remove()

Removes the first occurrence of a specified value.

Example:

```
numbers = [10, 20, 30, 20]
numbers.remove(20)
print(numbers)
```

Output:

```
[10, 30, 20]
```

5. pop()

Removes and returns an element. By default, it removes the last element.

Example:

```
numbers = [10, 20, 30, 40]
value = numbers.pop()
print(value)
print(numbers)
```

Output:

```
40  
[10, 20, 30]
```

6. clear()

Removes all elements from the list.

Example:

```
numbers = [10, 20, 30]  
numbers.clear()  
print(numbers)
```

Output:

```
[]
```

7. index()

Returns the index of the first occurrence of a specified value.

Example:

```
numbers = [10, 20, 30, 40]  
print(numbers.index(30))
```

Output:

```
2
```

8. count()

Returns the number of times a value occurs in the list.

Example:

```
numbers = [10, 20, 20, 30, 20]  
print(numbers.count(20))
```

Output:

```
3
```

9. sort()

Sorts the list in ascending order by default.

Example:

```
numbers = [40, 10, 30, 20]  
numbers.sort()  
print(numbers)
```

Output:

```
[10, 20, 30, 40]
```

10. reverse()

Reverses the order of elements in the list.

Example:

```
numbers = [10, 20, 30, 40]  
numbers.reverse()  
print(numbers)
```

Output:

```
[40, 30, 20, 10]
```

11. copy()

Creates a shallow copy of the list.

Example:

```
numbers = [10, 20, 30]
new_numbers = numbers.copy()

print(new_numbers)
```

Output:

```
[10, 20, 30]
```

Quick Revision

Method	Purpose
append()	Add one element at the end
extend()	Add multiple elements
insert()	Insert at a specific index
remove()	Remove a specific value
pop()	Remove and return an element
clear()	Remove all elements
index()	Find the index of a value
count()	Count occurrences
sort()	Sort the list
reverse()	Reverse the list
copy()	Create a copy of the list

Interview focus: append(), extend(), insert(), remove(), pop(), index(), count(), sort(), and reverse().